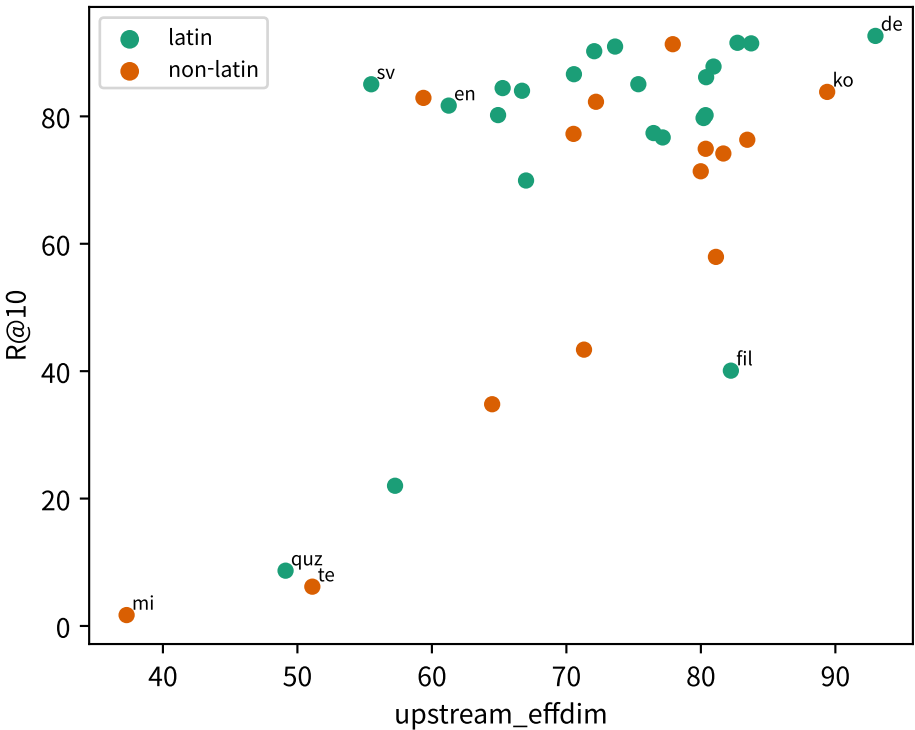
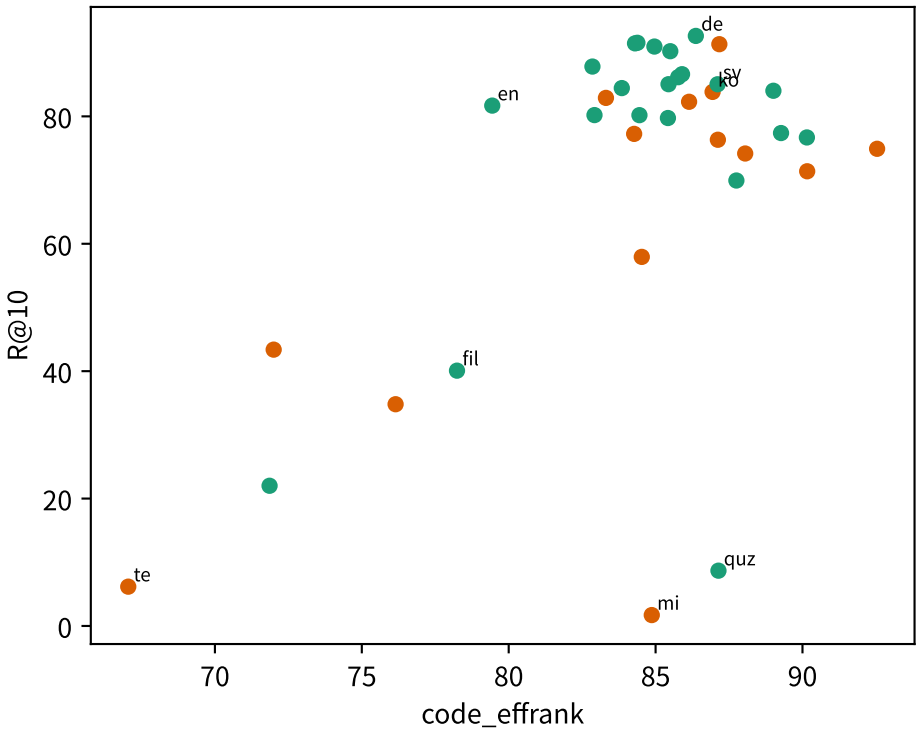


Part 1: candidate predictors of binary R@10 across 36 languages (XM3600, ceiling path)

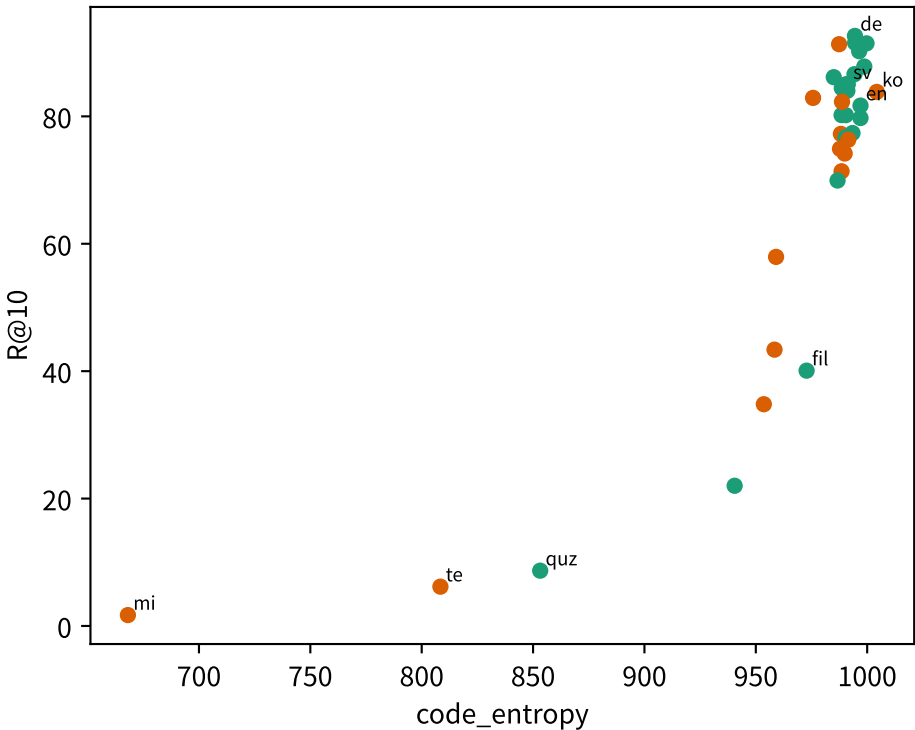
upstream text-emb eff-dim (actionable)
Spearman $\rho = +0.376$



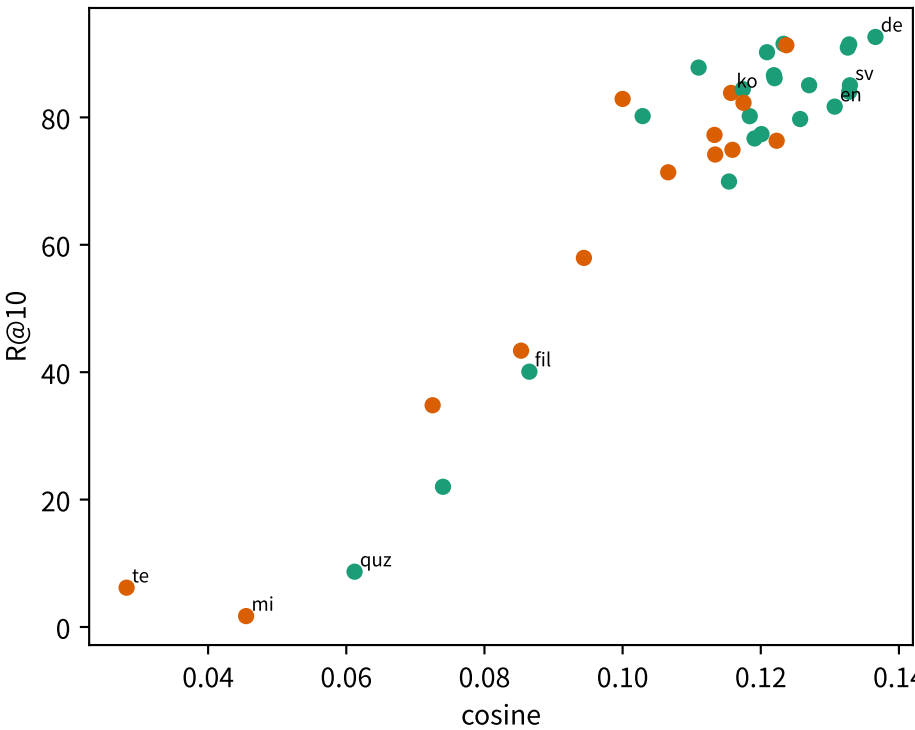
code effective-rank (downstream)
Spearman $\rho = +0.121$



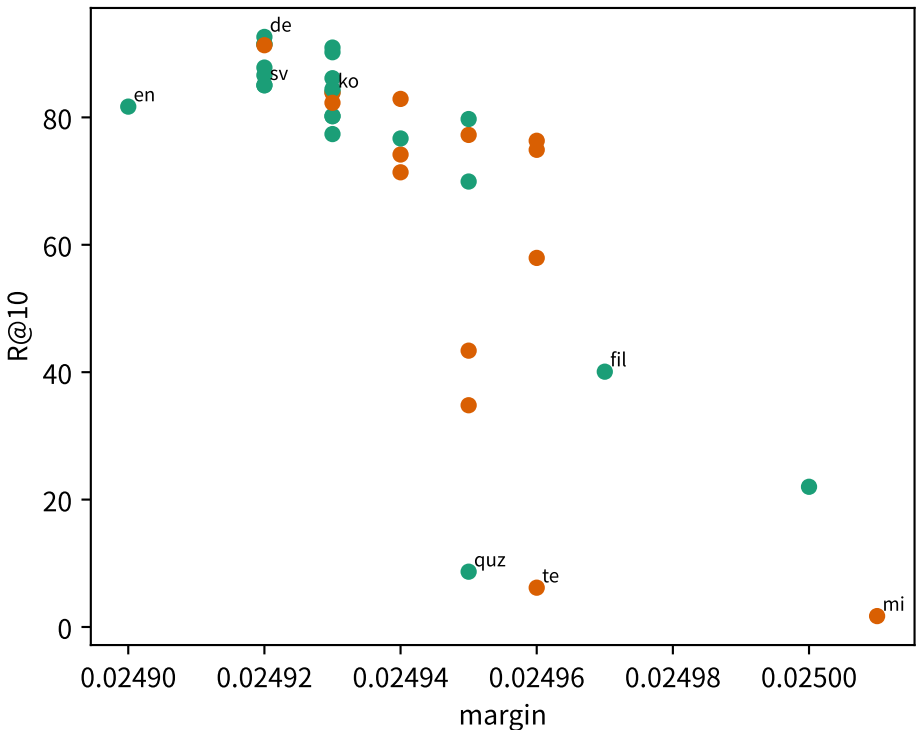
code bit-entropy (downstream/tautology-risk)
Spearman $\rho = +0.729$



pair cosine (proxy, embedding)
Spearman $\rho = +0.788$



mean|z| (proxy; ~constant)
Spearman $\rho = -0.866$



code dir-align (proxy; pair-Hamming = tautology)
Spearman $\rho = +0.980$

